

RNP

$$\mathcal{L}(S) \cap \mathcal{N} = \{0\}$$

$$0 \neq \theta^* \in \mathcal{N}$$

$\Rightarrow$
?

$$\theta^* \notin \mathcal{L}(S)$$

$$\mathcal{L}(S) = \{ \Delta \mid \|\Delta_S\|_1 \geq \|\Delta_{S^c}\|_1 \}$$

$$\|\theta_S^*\|_1 < \|\theta_{S^c}^*\|_1 \quad *$$

MIC

$\Rightarrow$
?

$$X = \begin{bmatrix}$$

$\left. \vphantom{\begin{bmatrix} \\ \\ \\ \end{bmatrix}} \right]$

$\rightsquigarrow$

$$\begin{array}{c} ? \\ S \\ \hline S^c \end{array} \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

MIC: $\left\| \frac{X^T X}{n} - I \right\|_\infty \leq \frac{1}{3s} \delta$

$\frac{1}{\sqrt{d}} \|v\|_1 \leq \|v\|_2 \leq \|v\|_1$

$\|v\|_\infty = \max_i |v_i|$

$|\langle \underline{x}, \underline{y} \rangle| \leq \|v\|_1 \cdot \underbrace{\|\underline{x}\| \cdot \|\underline{y}\|}_{\sqrt{d}}$

$i \left[\begin{matrix} -1 \\ \vdots \\ \langle x_i, x_j \rangle \\ \vdots \\ -1 \end{matrix} \right]_j$

$v^T Q v$

$= \left| \sum_{i,j} v_i v_j Q_{ij} \right|$

$\leq \underbrace{\max_{ij} |Q_{ij}|}_{\|Q\|_\infty} \underbrace{\sum_{i,j} |v_i| |v_j|}_{\|v\|_1^2}$

$\leq \left| \theta_s^{*T} \left[\frac{X^T X}{n} - I \right] \theta_s^* \right| \leq$

$\left\| \frac{X^T X}{n} - I \right\|_\infty \|\theta_s^*\|^2 \leq$

$\delta \|\theta_s^*\|^2 \leq \delta (\sqrt{s} \|\theta_s^*\|_2)^2$

$\delta s \|\theta_s^*\|_2^2$

$$\theta_s^{*T} \left(I - \frac{X^T X}{n} \right) \theta_s^* \leq \left| \theta_s^* \left(\frac{X^T X}{n} - I \right) \theta_s^* \right| \leq s\delta \|\theta_s^*\|_2^2$$

$$\|\theta_s^*\|_2^2 - \underbrace{\theta_s^{*T} \frac{X^T X}{n} \theta_s^*}_{\leq s\delta \|\theta_s^*\|_2^2} \leq s\delta \|\theta_s^*\|_2^2$$

$$\begin{aligned} \left(\frac{1-s\delta}{s\delta} \right) \|\theta_s^*\|_2^2 &\leq (1-s\delta) \|\theta_s^*\|_2^2 \leq \underbrace{\theta_s^{*T} \frac{X^T X}{n} \theta_s^*}_{\leq \left| \theta_s^{*T} \frac{X^T X}{n} \theta_s^* \right|} \leq \left| \theta_s^{*T} \frac{X^T X}{n} \theta_s^* \right| \\ &= \left| \theta_s^{*T} \left(\frac{X^T X}{n} - I \right) \theta_s^* \right| \\ &\leq \left\| \frac{X^T X}{n} - I \right\|_\infty \|\theta_s^*\|, \|\theta_{s^c}^*\|, \\ &\leq \cancel{s\delta} \|\theta_s^*\| \cdot \|\theta_{s^c}^*\|, \end{aligned}$$

$$X \theta^* = \underline{0}$$

$$[\theta_s^* + \theta_{s^c}^*] = \underline{0} \Rightarrow X \theta_s^* = -X \theta_{s^c}^*$$

$$\|\theta_s^*\|_1 \leq \underbrace{\frac{s\delta}{1-s\delta}}_{(1-\epsilon)} \|\theta_{s^c}^*\|_1$$

Suff.
Cond.

$$\frac{s\delta}{1-s\delta} \leq \underbrace{\frac{1}{2}}_{(1-\epsilon)}$$

for

$$\|\theta_s^*\|_1 \leq \|\theta_{s^c}^*\|_1$$

$$2s\delta \leq 1-s\delta \Rightarrow 3s\delta \leq 1 \Rightarrow \boxed{s \leq \frac{1}{3\delta}}$$

RIP

$X \Theta$

$$\begin{bmatrix} | & & | \\ x_1 & \dots & x_d \\ | & & | \end{bmatrix} \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_d \end{bmatrix} = \theta_1 x_1 + \dots + \theta_d x_d$$

$$(1-s) \|\theta^*\|_2 \leq \|X_S \theta^*\|_2 \leq (1+s) \|\theta^*\|_2$$